

HOPE

ResidentConnect

Team 3

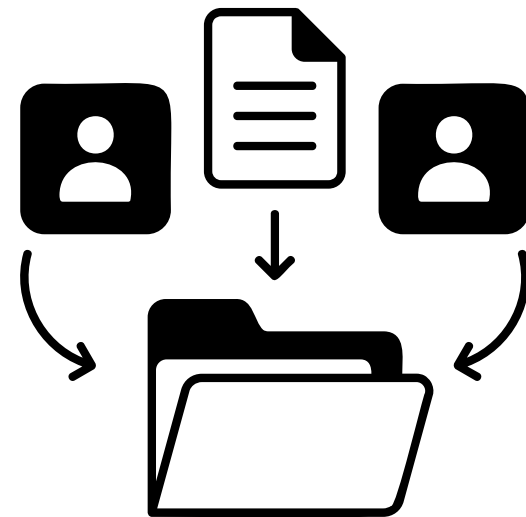
Jia Ying, Zheng Jing, Han-ying, Kai Xuan, Wei Xuan and Matthew

Executive Summary

Problem Statement:

New Home Community Service struggles with **fair distribution of limited resources** between tenants due to **lack of retrievable and structured data**.

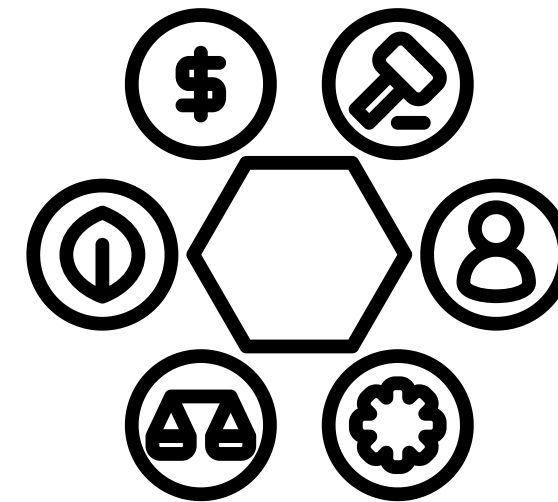
Painpoints:



Using excel to
track caselogs



Biased Observation



No deciding factor

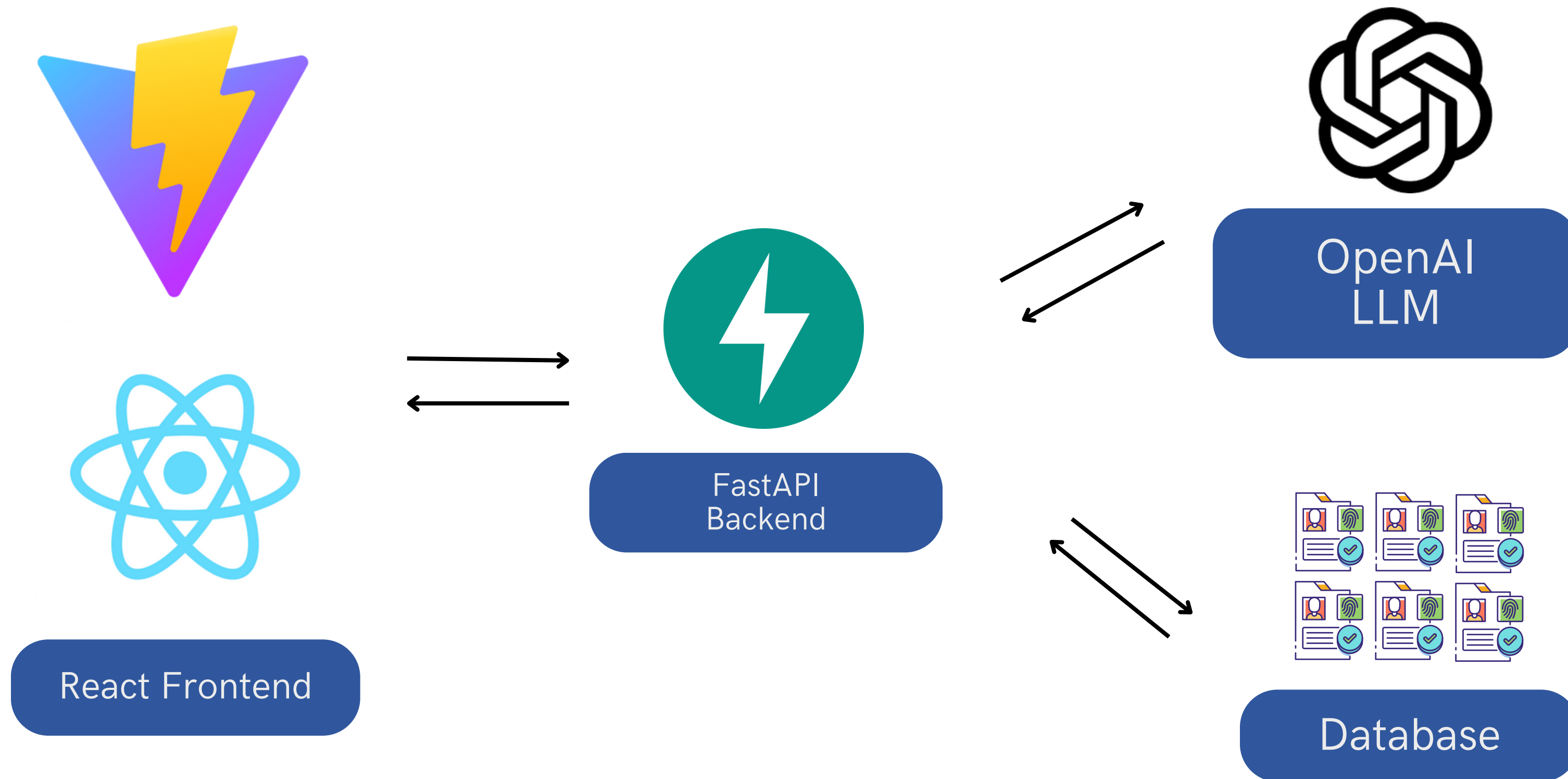
Key Features in our solution:

1. Tenant Profile → Collect structured data for more accurate recommendations for resource allocation
2. RAG Assisted Recommendation → Process data with set deciding factors, unbiased recommendations

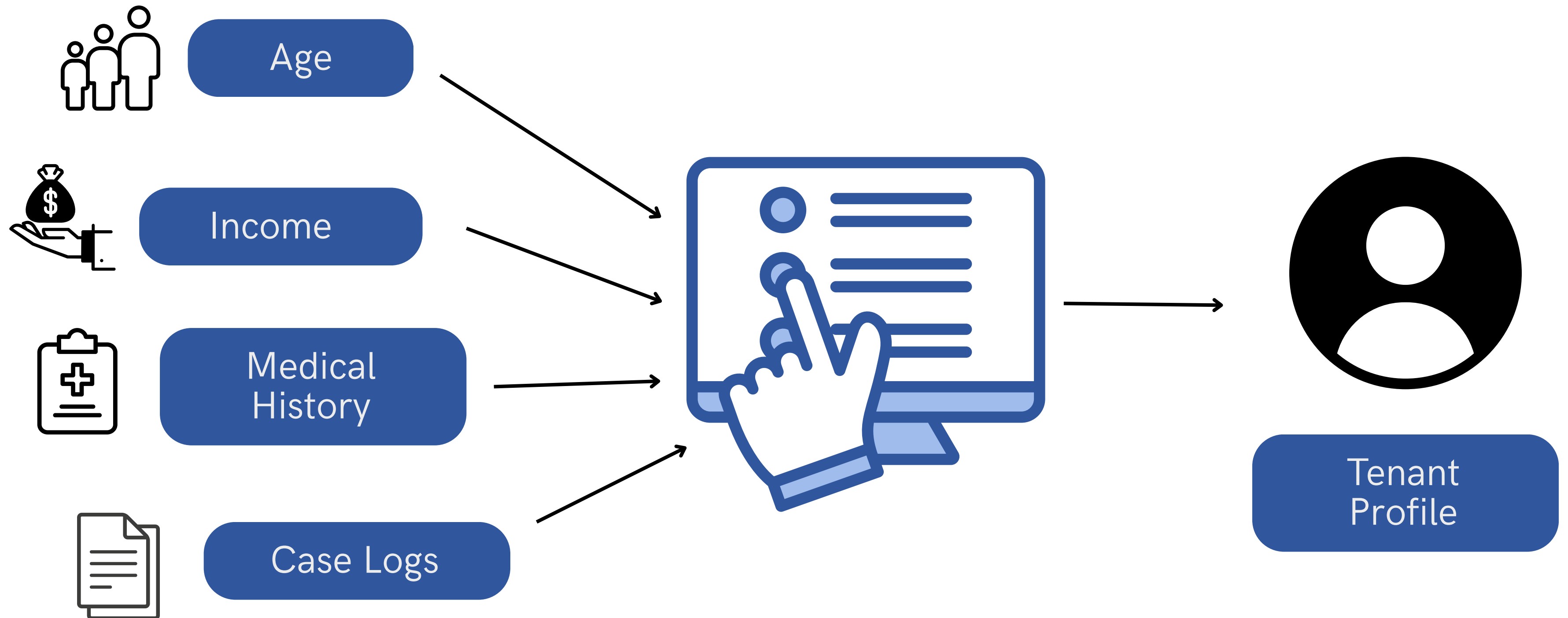
How we tackled the problem

Problem	Solution
Unclear tenant identities	Tenant Profile
Uncertainty about who should receive resources No visibility into tenants who haven't received resources recently	RAG Assisted Recommendation

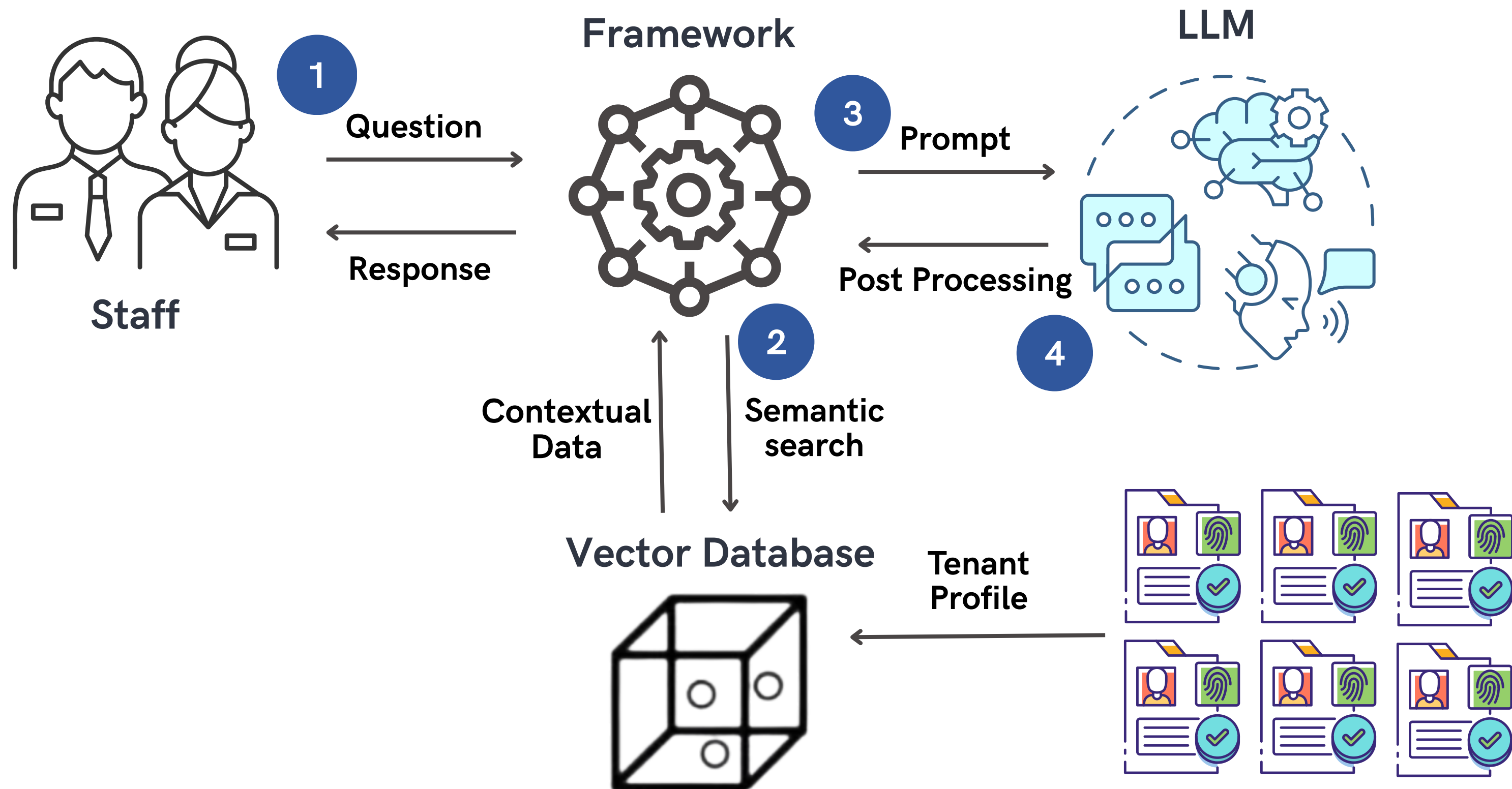
System Architecture



Tenant profiles



RAG-Assisted Recommendation



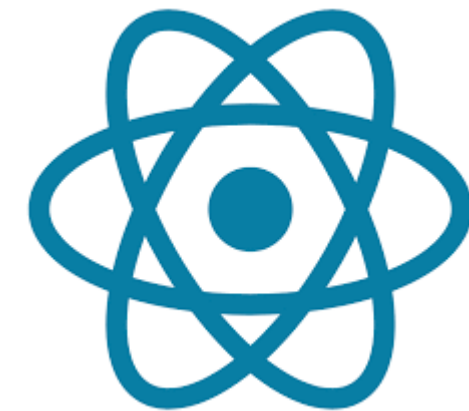
Web Application Architecture

Language: TypeScript

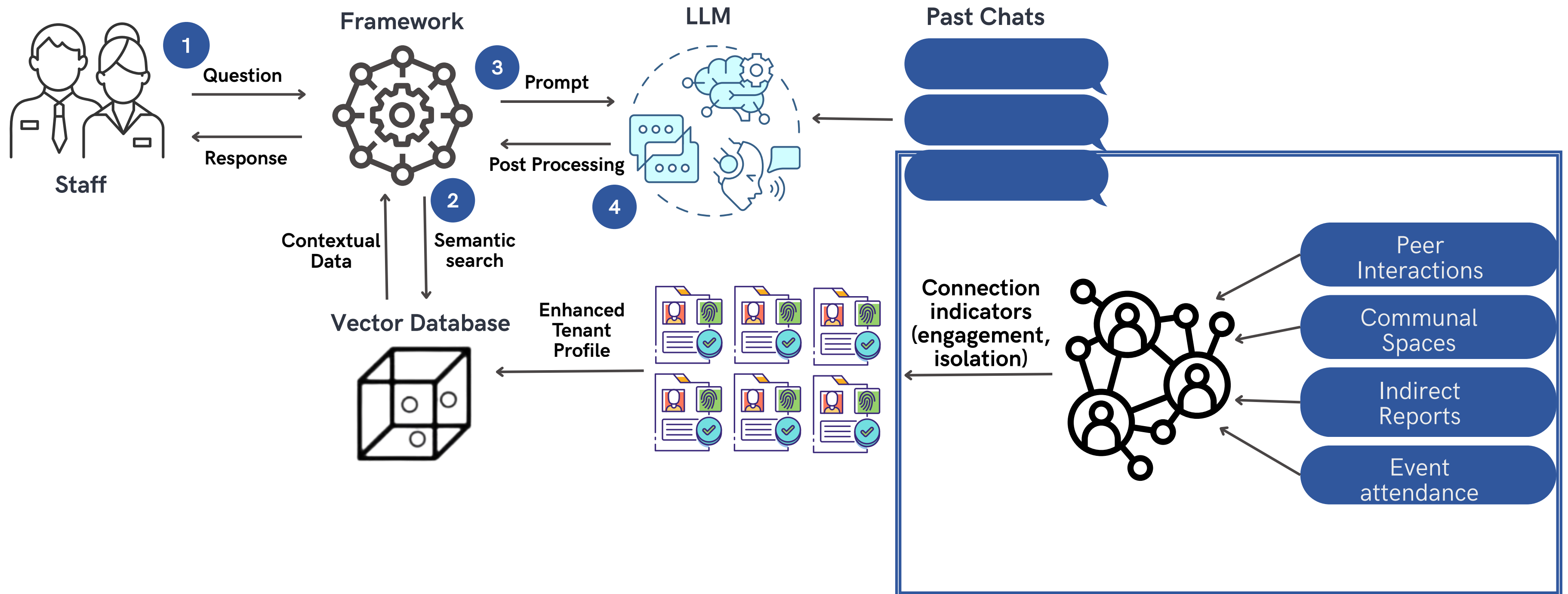
Framework: React

Router: TanStack Router

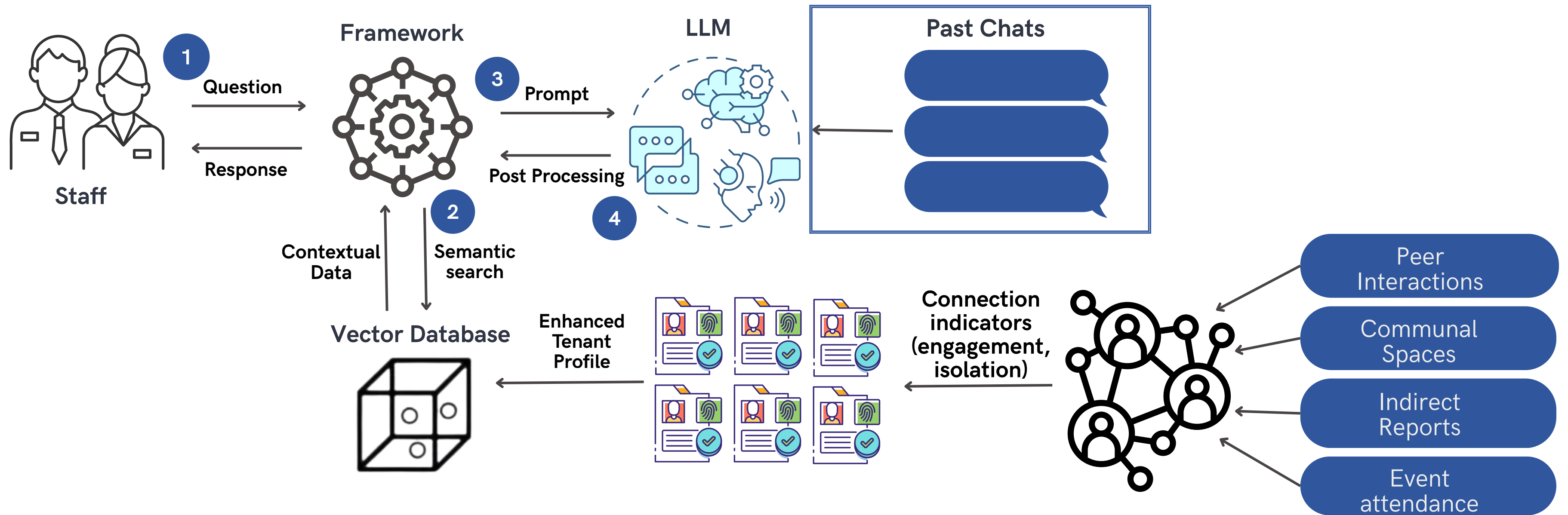
Styling/UI: Tailwind CSS + shadcn/ui + Lucide icons



Future Work

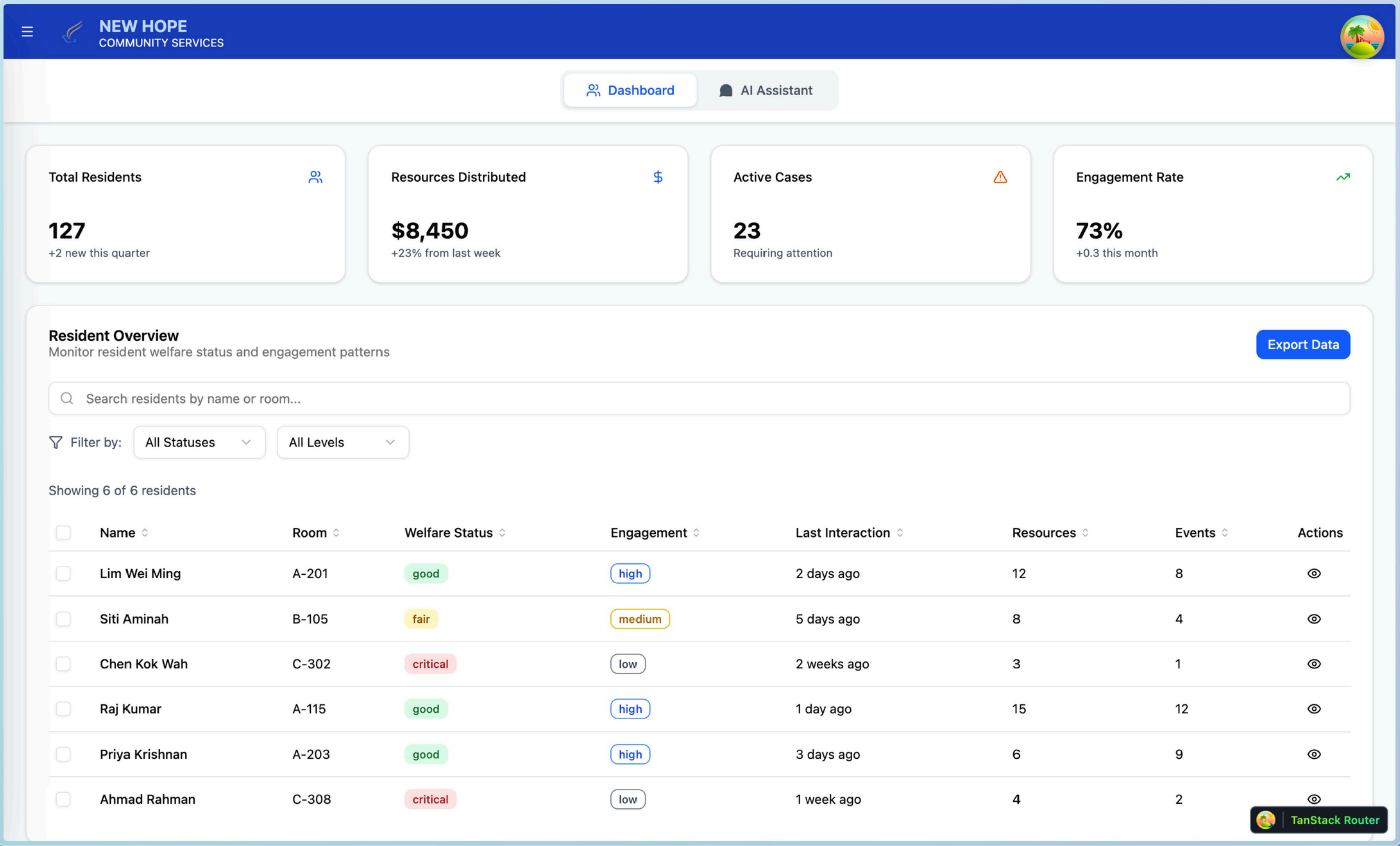


Future Work



Thank You!

Dashboard Page



Tenant Registration Form

**NEW HOPE**
COMMUNITY SERVICES

Tenant Registration

Full Name *	Location
Age	Marital Status
0	
Gender	Family Size
	0
Job Title	Industry
Annual Income (SGD)	Years of Experience
Select a range...	0

Case Logs

Add Log Entry

Clear

Save Profile

 TanStack Router Captured with Arc

Item Request Form



NEW HOPE
COMMUNITY SERVICES



Item Request

Person Name *

Date Needed By *

Description of Needs *

Other Notes

Clear

Submit Request



TanStack Router

AI Resource Assistant - General Chat

Dashboard

AI Assistant



AI-Powered Resource Assistant

Ask questions about residents and get AI-powered recommendations

General Chat

Resource Allocation

Hello! I'm your AI assistant for resident welfare management. You can ask me questions like "Who needs food assistance?" or "Show me residents with low engagement levels."

Quick Questions

Who needs immediate help?

People with health conditions

Family conflicts and mediation

Financial assistance analysis

Ask about residents, resources, or get recommendations...

Send

 TanStack Router

AI Resource Assistant - Resource Allocation

NEW HOPE
COMMUNITY SERVICES



Dashboard

AI Assistant

AI-Powered Resource Assistant

Ask questions about residents and get AI-powered recommendations

General ChatResource Allocation

⚙️ Resource Allocation

Basic Information

Resource Name

e.g., Emergency Food Assistance

Available Quantity

1

Description

Describe what this resource provides...

Eligibility Criteria

Who is eligible for this resource?

🗨️ Resource Allocation Mode: Fill out the form on the left to generate resource allocation recommendations. You can then ask follow-up questions about the recommendations.

Ask follow-up questions about the recommendations...

Send

 TanStack Router

 Captured with Arc

AI Resource Assistant - Resource Allocation Customisation

NEW HOPE
COMMUNITY SERVICES

Dashboard

AI Assistant

AI-Powered Resource Assistant

Ask questions about residents and get AI-powered recommendations

General Chat

Resource Allocation

Resource Allocation

Priority Weights

Recent Recipients

30%

Need Level

70%

Weight name

0

+

Additional Filters

Filter name

Filter value

+

Generate Refined Question

Job Title: Agricultural Engineer

Industry: Agriculture

Years of Experience: 23

Income Level: High

Health Information:

Health Conditions: Osteoporosis, ADHD

Lifestyle Factors: Vegetarian Diet

2. **Ava Smith**

- **Demographics:**

- Age: 32 years old

- Gender: Female

- Location: Washington, DC

- Marital Status: Divorced

- Family Size: 1 member

Ask follow-up questions about the recommendations...

Send

TanStack Router